

20. Which of the following cities is known as the oil capital of Europe?

- (a) Belfast (b) Aberdeen
(c) Leeds (d) Liverpool

U.P. P.C.S. (Pre) 2017

Ans. (b)

Aberdeen is often referred to as the Oil Capital of Europe. The petroleum industry in Aberdeen grew with the discovery of significant oil deposits in the North Sea in the mid-nineteenth century.

21. Which of the following Island is known as "The Land of Promise"?

- (a) Cuba (b) Java
(c) Sulawesi (d) Mindanao

U.P.R.O./A.R.O. (Pre) 2021

Ans. (d)

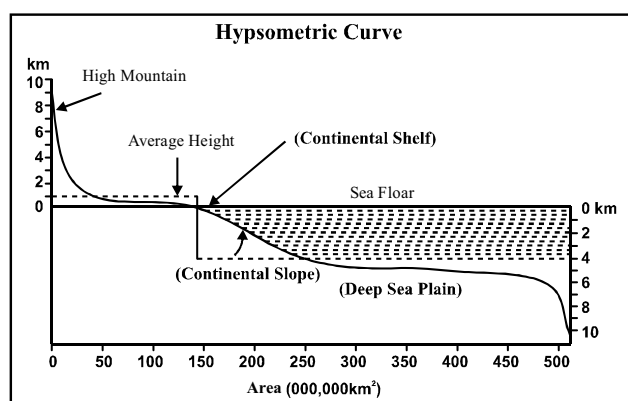
Mindanao is known as 'The Land of Promise'. Mindanao is an island in the Philippines archipelago.

The Hydrosphere

*The Hydrosphere is the total volume of water on the surface of our planet. Although the Earth's Hydrosphere has been around for longer than 4 billion years, it continues to change and rise. This is caused by seafloor spreading and continental drift. Hydrosphere occupies more than 2/3 of the total surface of the Earth. *Hydrosphere covers about 71% (70.9%) of Earth's surface. There is only one world ocean but the continents divide the ocean into five largest distinct parts -

- (i) Pacific Ocean (155557000 sq. km)
- (ii) Atlantic Ocean (76762000 sq. km)
- (iii) Indian Ocean (68556000 sq. km)
- (iv) Southern Ocean (20327000 sq. km)
- (v) Arctic Ocean (14056000 sq. km)

The Pacific Ocean is the largest in the world.



*The Oceanic Relief has more diversity than Lithosphere Relief.

*The Hypsographic Curve is a scientific way of describing the topography of the seafloor. We read the hypsographic curve like a graph with an x-axis (horizontal) and a y-axis (vertical). The x-axis tells us the elevation of the province. The curved line on the graph tells us about the provinces - where we see a change in the slope of the line, we are looking at a different province.

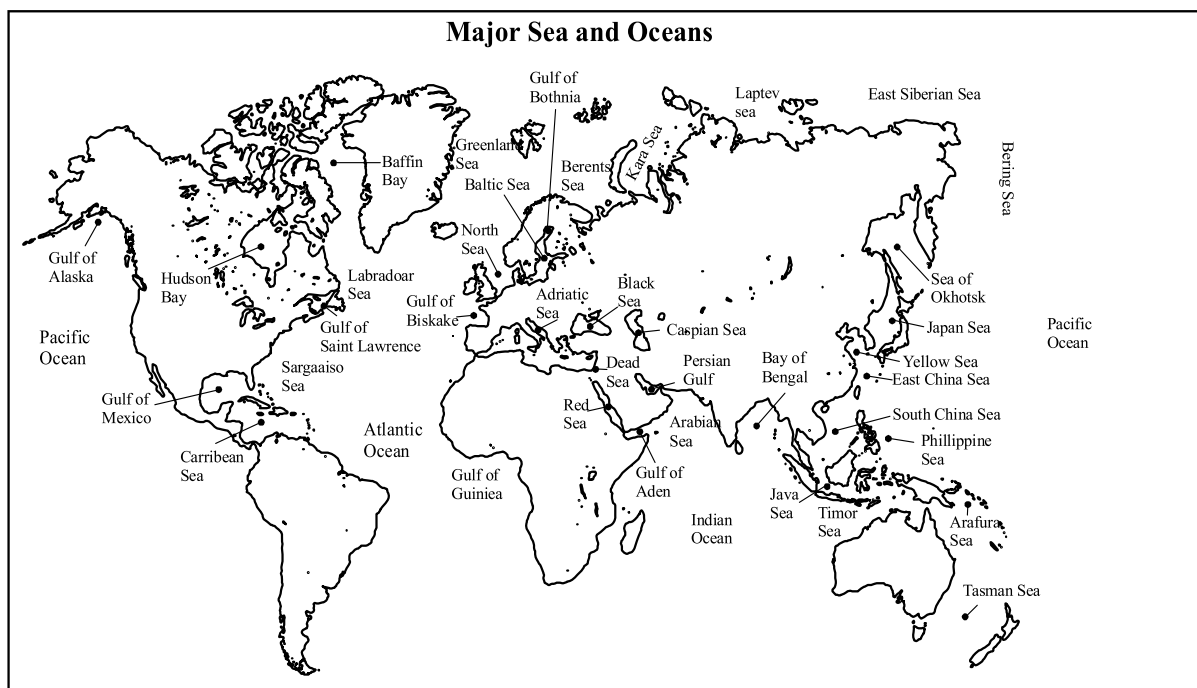
*The oceanic crust differs strikingly from the continental crust with respect to rock types, structure, landforms, age and origin. Four major divisions in the ocean relief are-

1. **Continental Shelf** - It is the submerged margin of continental mass extending from the shore to the first prominent slope.
2. **Continental Slope** - The slope that extends from the continental shelf down to the ocean deep.
3. **Continental Rise** - This is the gently sloping seafloor lying at the foot of the continental slope leading gradually into the Abyssal Plain.
4. **Abyssal Floor** - This is the large expanse of very smooth ocean floor found at depths of 3000 to 6000 meters. In fact they are the most widespread landforms on the Earth and occupy almost 76% of the total oceanic area.

***Oceanic trenches** are narrow, deep depression in the seafloor representing the line of subduction of an oceanic lithosphere plate. The ocean trenches are the lowest areas on Earth's surface and occupy almost 7% of the ocean's relief. They have very steep slopes. Examples of oceanic trenches include the Mariana Trench (Challenger Deep), Tonga and Kuril-Kamchatka, etc. Other important oceanic reliefs are -

- (i) **Oceanic Ridge** - formed by lava, that erupts beneath the sea and cools to form solid rocks.
- (ii) **Mid Oceanic Ridge** - an underwater mountain system formed by plate tectonics.
- (iii) **Guyot** - A Guyot, also known as a tablemount is an isolated underwater volcanic mountain with a flat top below the surface of the sea. They are commonly found in the Pacific Ocean, but they have been identified in all the oceans except the Arctic Ocean.
- (iv) **Submarine Canyon** - Narrow steep-sided valleys that cut into continental slopes and continental rises of the ocean.
- (v) **Fracture Zone** - A linear oceanic feature often hundreds, even thousands of kilometers long. They are a consequence of plate tectonics.

Similarly, other ocean relief features include island arcs, atolls, coral reefs, submerged volcanoes, sea-scarps etc.



*The **Mid Atlantic Ridge** is a mid-ocean ridge, a divergent tectonic plate or constructive plate boundary located on the floor of the Atlantic Ocean and part of the most extensive mountain range in the world. In the North Atlantic it separates the Eurasian and North American plates and in South Atlantic it separates the African & South American plates.

*The **Ninety East Ridge** is a ridge on the Indian Ocean floor named for its near by parallel strike along the 90th meridian at the center of the Eastern Hemisphere.

*Examples of such islands in the Indian Ocean are - Laccadive, Maldives, Chagos, New Amsterdam, Saint Paul, Kerguelen, etc.

***Datum line** is the horizontal line to sea level used to measure heights and depths.

*96.5% of the total water present in the ocean is saline in nature. Out of the remaining 2.5% is fresh water. 1.7% are in the form of glaciers and polar ice. Hence, less than 1% of fresh water resources is available for use.

*Out of the total available fresh water, 68.7% is inaccessible as it is in the form of glaciers and ice caps. 30.1% is in the form of groundwater, 0.3% is in the form of surface water and only 0.9% is in the form of other available sources.

1. With reference to Ocean Mean Temperature (OMT), which of the following statements is/are correct?

1. OMT is measured up to a depth of 26°C isotherm which is 129 meters in the south-western Indian Ocean during January-March.

2. OMT collected during January-March can be used in assessing whether the amount of rainfall in monsoon will be less or more than a certain long-term mean.

Select the correct answer using the code given below:

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2020

Ans. (b)

The 26°C isotherm is noticed at depths varying between 50-100 meters. During January-March, the mean 26°C isotherm in the south western Indian Ocean is at a depth of 59 meters. Ocean Mean Temperature (OMT) compared to sea surface temperature (SST) better predicts whether the total amount of rainfall that India will receive during monsoon season will be less or more than the long term mean of 887.5 mm.

2. The surface of the earth covered with water is approximately?

- (a) One-fourth (b) Half
(c) Two-thirds (d) Three-fifth (3/5)

41st B.P.S.C. (Pre) 1996

Ans. (c)

Around 29% (29.1%) of the total surface area of the Earth is land (about 148172000 sq. km) and 71% (70.9%) (about 361900000 sq. km) is covered by ocean. Hence, the surface of Earth covered with water is approximately more than two-thirds of the total surface area.

3. On the sea level, the nearest place to the center of earth is ?

- (a) North Pole (b) Tropic of Capricorn
(c) Tropic of Cancer (d) Tropic of Equator

38th B.P.S.C. (Pre) 1992

Ans. (a)

The nearest point to the center of the Earth lies near the Arctic Ocean (at a depth of 4 km from the surface) which is the 6353 km from the center of the Earth.

4. Datum line :

- (a) Is a horizontal line to the sea level from where heights and depths are measured
(b) Is the mean of primary and secondary data
(c) International date line
(d) Is an imaginary line that passes through zero degree meridian

U.P.P.C.S. (Pre) 1991

U.P.P.C.S. (Pre) 1994

Ans. (a)

The datum line is a horizontal line to sea level from which all elevations and depths are measured.

5. The part of our hydrosphere is –

- (a) Atlantic Ocean (b) Indian Ocean
(c) Pacific Ocean (d) Antarctica Ocean

Uttarakhand P.C.S. (Pre) 2010

Ans. (c)

The area of given oceans is as follows – Pacific Ocean (155557000 sq. km), Atlantic Ocean (76762000 sq. km), Indian Ocean (68556000 sq. km), Southern Ocean (20327000 sq. km) and Arctic Ocean (14056000 sq. km). Hence largest part of our hydrosphere is the Pacific Ocean.

6. A ridge of 64000 km. length and 2000 km to 2400 km width passing through the north and south Atlantic oceanic basin enter into the south pacific oceanic basin through Indian oceanic and then from the middle of Australia and Antarctic. This ridge is –

- (a) Socotra-Lakshadweep - Changos-Ridge
(b) Pacific - Antarctica Ridge
(c) Dofin - Challenger Ridge
(d) Mid - Oceanic Ridge

47th B.P.S.C. (Pre) 2005

Ans. (d)

Above qualities are related to Mid Oceanic Ridge. Mid Oceanic Ridge encircles the globe with a length of approximately 40390 miles or 65000 kilometer, covering the Arctic, Atlantic, Indian and Pacific Oceans. This is the single largest geological phenomenon on the surface of Earth.

7. Where is 'Ninety East Ridge' situated?

- (a) Pacific Ocean (b) Indian Ocean
(c) Atlantic Ocean (d) Arctic Ocean

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Ninety East Ridge (90° E Ridge) is the name of a ridge located in the Indian Ocean. Located near 90° east longitude (meridian), this ridge is 5000 km long parallel to it and stretches from 34° south latitude to 10° north latitude with an average width of 200 km.

8. The tension of gravitational water in the ground is –

- (a) below 1/3 atmosphere (b) At 1.25 atmosphere
(c) At 5 atmosphere (d) At 15 atmosphere

U.P.U.D.A./L.D.A. (Spl) (Mains) 2010

Ans. (a)

Gravitational water occupies the larger soil pores (macro-pores) and moves down readily under the force of gravity. Gravitational water under the surface of soil remains in a state of zero or less than 1/3 of the atmosphere.

9. Consider the following statements:

1. On the planet Earth, the fresh water available for use amounts to about less than 1% of the total water found.
2. Of the total freshwater found on the planet Earth, 95% is bound up in polar ice caps and glaciers.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2010

Ans. (a)

96.5% of the total water available on Earth is salt water found in ocean. Out of the remaining 2.5% fresh water, 1.7% is in the form of glaciers and polar ice, which is unusable for us. So in this way total available fresh water to use is less than 1% of the total water available on Earth. Out of the total amount of fresh water (2.5%), 68.7% is found in the form of glaciers and polar ice, 30.1% in the form of underground water, 0.3% in the form of surface water and the remaining 0.9% is found in other forms. Hence only statement (1) is correct.